

Solar Energy

Pupil sheet

Name: _____

Date: _____

Safety: Never look at the sun. The box is not waterproof: bring it in if it rains.

1. What I see on the screen

Seven cards: **two measured by sensors, five worked out from them** after the word **Model:**. Tick **M** (measured) or **W** (worked out) — the most important question of the lesson.

The card	M	W	The card	M	W
Illuminance (lx)	☐	☐	Efficiency (%)	☐	☐
Air temp. (°C)	☐	☐	Power, 1 m ² (W)	☐	☐
Irradiance (W/m ²)	☐	☐	Energy, 1 m ² (Wh)	☐	☐
Panel temp. (°C)	☐	☐			

The screen holds only the last two minutes; the day's curve is in the file from **Export data**.

The station log — written once, at the start:

Place: _____

Orientation: _____

Tilt: _____

2. What I do on my own

First I predict; I do not change my prediction after I measure. Tilted by 60°, the sensor will show about _____ of what it showed facing the source squarely,

because _____

The tilt investigation. $\cos \theta$ is already filled in; you measure and work out the ratio. **At the end return to 0°:** if it differs much from the first reading, the sun changed and the series is not sound.

Angle θ	$\cos \theta$	Light (lx)	Ratio $L(\theta)/L(0)$
0°	1.00		1.00
30°	0.87		
60°	0.50		
75°	0.26		
0° (control)	1.00		

The sky while measuring (circle one): clear broken overcast

Two days, compared. The peak says who shone hardest for a moment. **The area says who produced more energy** — and only that matters to a panel.

	Day 1	Day 2	Which is larger?
Sky (one word)			
Peak light (lx)			
Time of the peak			
Area under the curve			

The area, by the trapezium rule: for each neighbouring pair of readings, their mean times the time between them; then add it all up.

3. The conclusion I write

I predicted that, and the data show that _____

At 60° the measured ratio does, or does not, match 0.50 because _____

The day that produced more energy is the one with the larger area, not the larger peak, because

Exit ticket

Which of the numbers on the screen were measured and which were worked out?
